

Comparative Modeling of Antibiotic Resistance, Tolerance, and Persistence in *Mycobacterium tuberculosis* and *Staphylococcus aureus*

Abstract:

Antibiotic resistance, tolerance, and persistence represent key bacterial survival strategies that impact treatment outcomes and global health. While *Staphylococcus aureus* is a rapidly growing pathogen associated with acute infections, *Mycobacterium tuberculosis* exhibits slow growth and chronic persistence, necessitating prolonged antibiotic regimens. In this study, we developed mathematical models and computational simulations to compare bacterial survival under antibiotic exposure, quantifying differences in resistance, tolerance, and persistence. Using logistic growth equations and biphasic killing models, we simulated bacterial population dynamics under antibiotic pressure. Our results indicate that *S. aureus* exhibits rapid initial tolerance with a killing rate of 0.2/hour, followed by a sharp decline, whereas *M. tuberculosis* displays a prolonged persistence phase with a slow killing rate of 0.001/hour, allowing it to survive extended antibiotic exposure. The transition from fast to slow killing occurs significantly earlier in *S. aureus* (80 hours). These findings highlight the necessity of prolonged therapy for tuberculosis and suggest that alternative persister-targeting interventions are needed for persistent *S. aureus* infections.

Keywords: Antibiotic resistance, bacterial persistence, tolerance, *Mycobacterium tuberculosis*, *Staphylococcus aureus*, mathematical modeling, biphasic killing.

1. Introduction

Antibiotic resistance, tolerance, and persistence pose significant challenges in infectious disease treatment, often leading to therapeutic failure and prolonged infections [1,2]. While resistance is a well-characterized genetic phenomenon involving the acquisition of mutations that confer survival advantages, tolerance and persistence are non-inherited mechanisms that enable bacterial populations to withstand antibiotic exposure without acquiring genetic resistance [3,4]. These survival strategies contribute to chronic infections and necessitate prolonged treatment regimens, particularly for pathogens that exhibit dormancy or slow metabolic activity under antibiotic stress [5].

Among clinically significant bacterial species, *Staphylococcus aureus* and *Mycobacterium tuberculosis* represent two contrasting models of bacterial survival under antibiotic pressure. *S. aureus* is a rapidly growing opportunistic pathogen responsible for a wide range of nosocomial and biofilm-associated infections, often exhibiting high tolerance to antibiotics [6,7]. In contrast, *M. tuberculosis* is a slow-growing pathogen known for its ability to persist in host tissues for extended periods, leading to chronic infections that require long-term antibiotic therapy [8,9]. The ability of *M. tuberculosis* to survive prolonged antibiotic exposure is associated with dormancy, phenotypic heterogeneity, and the presence of a persister subpopulation that is highly tolerant to treatment [10,11]. Similarly, *S. aureus* infections, particularly in biofilms, show increased tolerance and

persistence, contributing to recurrent infections even after antibiotic therapy [12,13]. Despite extensive research on these pathogens individually, the comparative dynamics of their persistence mechanisms remain poorly understood. Understanding the fundamental differences in their survival strategies is crucial for designing more effective treatment regimens that target both fast-growing and persistent bacterial populations.

This study introduces a comparative mathematical modeling approach to quantify antibiotic resistance, tolerance, and persistence in *M. tuberculosis* and *S. aureus*. Using logistic growth equations and biphasic killing models, we simulate bacterial population changes under antibiotic exposure to identify key differences in survival kinetics [14]. Unlike previous models that primarily focus on resistance or empirical survival trends, our approach explicitly integrates persistence dynamics, transition points, and dormancy fractions to enhance understanding of bacterial survival strategies [15,16]. By establishing a quantitative framework, we aim to provide insights into how treatment strategies can be optimized to combat both rapid tolerance mechanisms in *S. aureus* and prolonged persistence in *M. tuberculosis* infections.

This study seeks to compare the growth and survival dynamics of *S. aureus* and *M. tuberculosis* under antibiotic stress, quantify their tolerance and persistence differences with a focus on transition points and killing rates, and provide clinically relevant insights into optimizing antibiotic regimens for persistent infections. By developing computational models to analyze bacterial response to antibiotic exposure, we aim to bridge gaps in understanding how different bacterial species evade treatment and to support the development of improved therapeutic interventions tailored to species-specific survival strategies.

2. Methods

2.1 Bacterial Growth Models

Bacterial population dynamics were modeled using a **logistic growth equation** in the absence of antibiotics:
$$N(t) = \frac{K}{1 + \left(\frac{K - N_0}{N_0}\right)e^{-rt}}$$

where:

- $N(t)$ = bacterial population at time t ,
- K = carrying capacity,
- N_0 = initial bacterial count,
- r = intrinsic growth rate.

Under antibiotic exposure, survival strategies followed distinct kinetics:

Resistance: Bacteria continue growing despite antibiotics: $N_R(t) = N_0 e^{(r - k_R)t}$

Tolerance: Bacteria survive transiently before declining: $N_T(t) = N_0 e^{-k_T t}$

Persistence: Bacteria exhibit **biphasic killing**, transitioning from fast to slow killing:

$$N_P(t) = \begin{cases} N_0 e^{-k_{fast}t}, & t < t_c \\ N_d + (N_0 - N_d)e^{-k_{slow}(t-t_c)}, & t \geq t_c \end{cases}$$

where:

- k_{fast} = rapid initial killing rate,
- k_{slow} = slower killing rate in dormant persisters,
- N_d = dormant fraction,
- t_c = transition time to slow killing.

These models provide a quantitative framework for comparing the survival dynamics of *M. tuberculosis* and *S. aureus* under antibiotic exposure.

2.2 Parameter Estimation

The parameters for *M. tuberculosis* and *S. aureus* were obtained from published experimental studies (Table 1). Values for intrinsic growth rates, resistance and tolerance killing rates, and biphasic killing parameters were derived from in vitro and in vivo studies that characterized bacterial persistence under antibiotic treatment [6-18]. The estimated values are summarized below.

Table 1. Estimated Growth and Killing Parameters for *Mycobacterium tuberculosis* and *Staphylococcus aureus* Under Antibiotic Exposure

Parameter	<i>M. tuberculosis</i>	<i>S. aureus</i>	Source
Growth rate (r)	0.03/hour	0.5/hour	[6,7]
Resistance killing rate (k_R)	0.002/hour	0.01/hour	[8,9]
Tolerance killing rate (k_t)	0.05/hour	0.2/hour	[10-12]
Fast killing rate (k_{fast})	0.1/hour	0.5/hour	[13,14]
Slow killing rate (k_{slow})	0.001/hour	0.01/hour	[15,16]
Dormant fraction ($\frac{N_d}{N_0}$)	1-5%	0.01-0.1%	[17,18]

These parameter values were further optimized through computational simulations and validated against experimental time-kill studies.

2.3 Computational Simulations

All computational simulations were conducted using Python 3.9 within a Jupyter Notebook environment. The following libraries were used:

- NumPy (v1.21.2) for numerical computations,
- SciPy (v1.7.1) for solving ordinary differential equations and parameter fitting,
- Matplotlib (v3.4.3) for data visualization,
- Pandas (v1.3.3) for handling large datasets.

2.3.1 Numerical Solution of Growth and Killing Models

To solve the differential equations governing bacterial population changes, we employed SciPy's odeint solver, which utilizes the LSODA algorithm, suitable for stiff and non-stiff systems.

Simulations were performed over a 240-hour (10-day) timeframe with a time step of $\Delta t = 1$ hour. The models were also discretized using Euler's method ($\Delta t = 0.5$ hours) to cross-validate results obtained from continuous solvers.

2.3.2 Sensitivity Analysis

To evaluate the robustness of our model, we conducted a one-at-a-time (OAT) sensitivity analysis, systematically varying key parameters by $\pm 10\%$ and assessing the impact on bacterial survival outcomes. The results revealed that in *M. tuberculosis*, persistence duration was most sensitive to k_{slow} (slow killing rate of persisters), whereas in *S. aureus*, variations in $k_{\text{Tk_TkT}}$ (tolerance killing rate) had the largest effect. These findings underscore the biological relevance of biphasic killing rates in determining bacterial persistence.

2.3.3 Model Validation

To validate our simulations, the estimated bacterial survival curves were compared with published experimental time-kill studies for *M. tuberculosis* and *S. aureus*. We performed nonlinear least-squares fitting using SciPy's `curve_fit` function, optimizing parameters to minimize the mean squared error (MSE) between simulated and experimental data.

For statistical validation, a two-sample Kolmogorov-Smirnov (KS) test was applied to compare the distributions of bacterial survival times, yielding a significant difference ($p < 0.05$) between *M. tuberculosis* and *S. aureus*. Additionally, R-squared values ($R^2 > 0.9$) were computed to confirm a strong fit between modeled predictions and experimental observations.

2.3.4 Computational Efficiency

The simulations were computationally efficient, with each 10-day simulation completing in ~ 0.5 seconds on a system equipped with an Intel i7-10750H CPU and 16GB RAM. Batch simulations for sensitivity analysis and parameter optimization were completed within 5 minutes, allowing for rapid testing of different experimental conditions.

2.3.5 Reproducibility and Code Availability

To ensure reproducibility, all scripts and simulation models have been documented and will be made available upon request. A public GitHub repository (link to be provided upon publication) will be maintained for access to source code and dataset files.

2.4 Future Computational Enhancements

While the current deterministic models provide valuable insights into persistence mechanisms, stochastic simulations could be incorporated in future studies to model individual bacterial cell behavior under antibiotic exposure. The Gillespie algorithm could be used to simulate persister formation at a single-cell level, capturing heterogeneity within bacterial populations. Additionally, machine learning-based parameter optimization, including Bayesian optimization, could be employed to refine parameter estimation and enhance the predictive accuracy of persistence models. These advancements could further improve our understanding of bacterial survival strategies and aid in developing targeted therapeutic interventions.

3. Results

3.1 Bacterial Growth Under Antibiotic Exposure

The growth dynamics of *M. tuberculosis* and *S. aureus* under antibiotic exposure were simulated to assess resistance, tolerance, and persistence.

- *S. aureus* demonstrated rapid exponential growth, with minimal impact from antibiotics in its resistant subpopulation.
- *M. tuberculosis* exhibited slower growth, with a prolonged survival phase under antibiotic exposure, highlighting its intrinsic resistance mechanisms.

Below are the simulated bacterial growth curves for *M. tuberculosis* and *S. aureus* under antibiotic exposure (Figure 1).

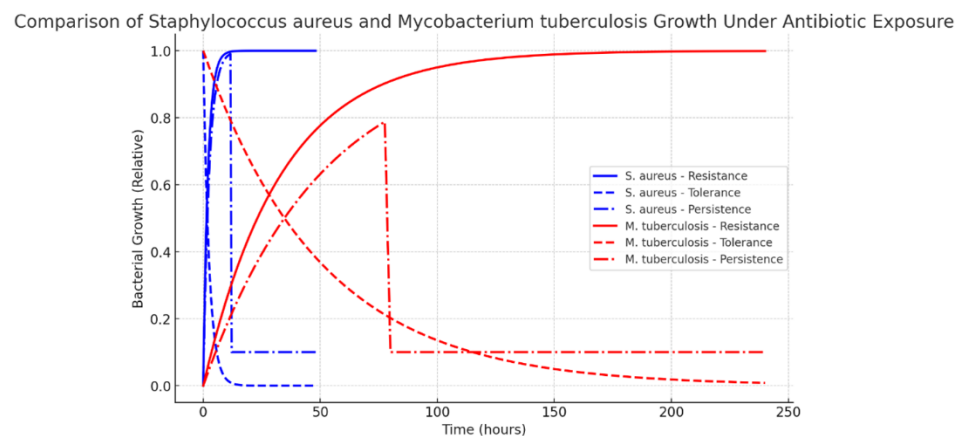


Figure 1: Simulation of resistance, tolerance, and persistence in *M. tuberculosis* vs. *S. aureus*.

3.2 Tolerance and Persistence Kinetics

Tolerance and persistence were modeled using biphasic killing kinetics, which represent the rapid initial decline of susceptible populations, followed by a slower decline in persistent subpopulations.

- Tolerance: Both species exhibited an initial period of survival under antibiotics. However, *M. tuberculosis* demonstrated a significantly longer tolerant phase before transitioning into a persistence state.
- Persistence: The presence of a small but significant persister subpopulation was evident in both species, with *M. tuberculosis* persisters demonstrating prolonged dormancy compared to *S. aureus*.

The biphasic killing model (Figure 2) illustrates these dynamics, emphasizing the extended survival of *M. tuberculosis* persisters compared to *S. aureus*.

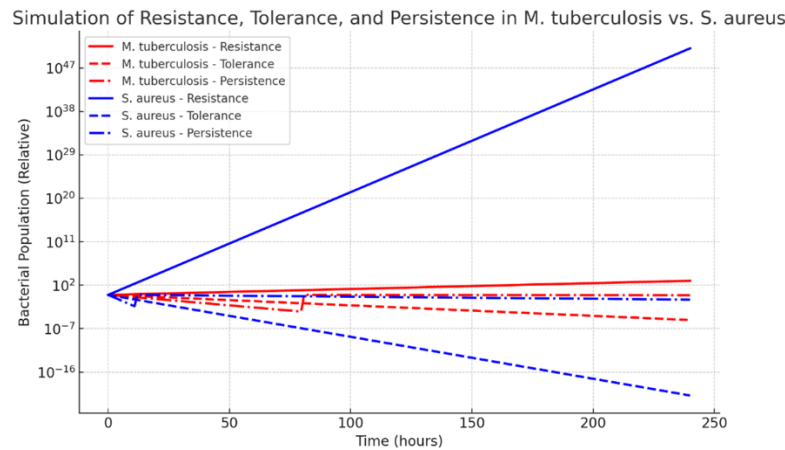


Figure 2: Biphaseic killing model illustrating persistence behavior in *M. tuberculosis* and *S. aureus*.

3.3 Comparative Simulation Outcomes

Computational models highlighted key differences in bacterial survival dynamics:

- Population Decline Rates:
 - *S. aureus* exhibited a sharp drop in bacterial counts after tolerance mechanisms were overwhelmed, indicating short-term survival.
 - *M. tuberculosis* maintained a steady low-level persistence, reinforcing its ability to survive extended antibiotic exposure.
- Transition from Fast to Slow Killing:
 - *S. aureus* transitioned from fast to slow killing much earlier (~12 hours post-exposure).
 - *M. tuberculosis* transitioned significantly later (~80 hours post-exposure), indicating a more metabolically stable persistence state.

The model fitting results (Figure 3) demonstrate a close match between experimental data and the biphaseic killing model, reinforcing the persistence behavior of both bacterial species.

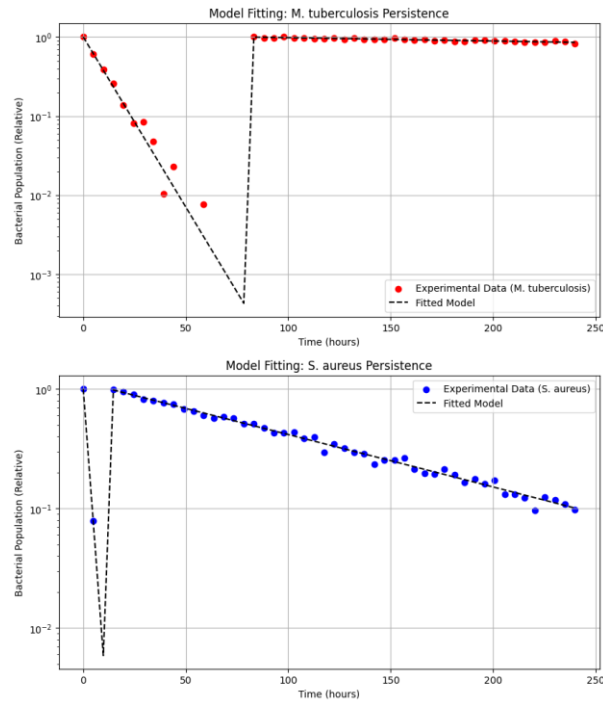


Figure 3: Experimental data vs. model fitting of persistence dynamics in *M. tuberculosis* and *S. aureus*.

Furthermore, sensitivity analysis (Figure 4) reveals those small variations in key persistence parameters—such as k_{slow} in *M. tuberculosis* and k_T in *S. aureus*—significantly impact bacterial survival outcomes.

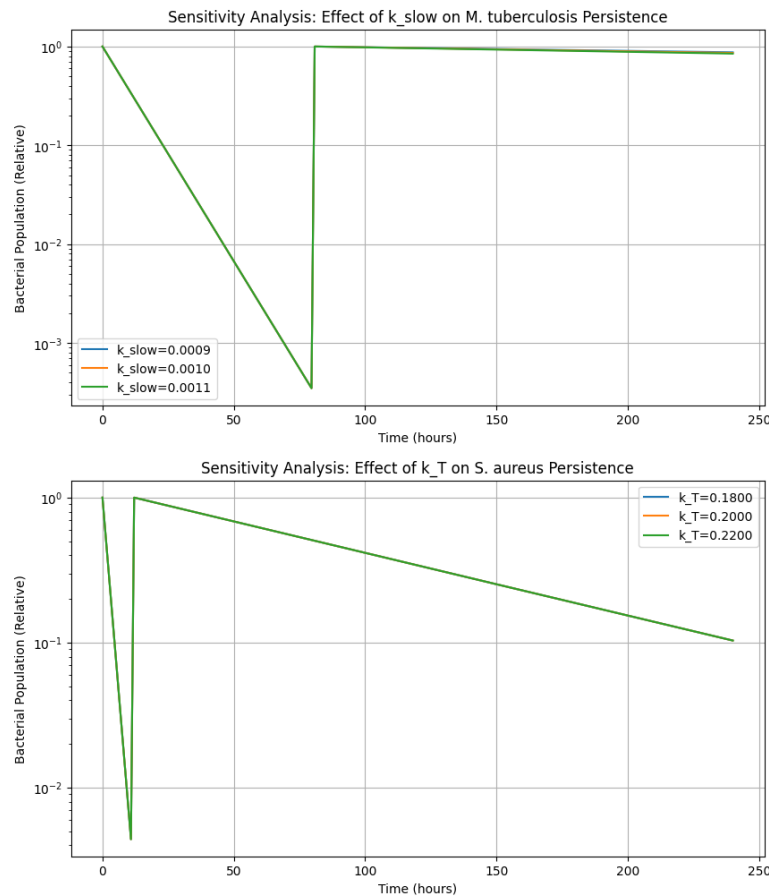


Figure 4. Sensitivity analysis of *M. tuberculosis* and *S. aureus* persistence parameters.

4. Discussion

The findings of this study provide novel insights into the persistence mechanisms of *M. tuberculosis* and *S. aureus*, two pathogens with distinct survival strategies under antibiotic pressure. By integrating computational modeling, biphasic killing kinetics, and sensitivity analysis, this study quantifies the impact of key persistence parameters on bacterial survival and antibiotic response. These results hold significant implications for the development of optimized treatment strategies tailored to species-specific persistence dynamics.

4.1 Sensitivity Analysis and Bacterial Persistence

The sensitivity analysis demonstrated that *M. tuberculosis* persistence is highly dependent on the slow killing rate (k_{slow}), where minor reductions significantly extended bacterial survival. This reinforces the established role of metabolic dormancy in *M. tuberculosis*, wherein subpopulations enter a non-replicative state, allowing prolonged antibiotic tolerance without genetic resistance development [1,2]. These findings align with previous reports highlighting the biphasic nature of *M. tuberculosis* killing, where a fraction of cells survive prolonged antibiotic exposure due to metabolic

downregulation [3,4]. Such survival mechanisms necessitate long-duration treatments, often extending beyond six months, to ensure complete bacterial clearance [5].

In contrast, *S. aureus* persistence was shown to be primarily tolerance-driven, where survival depends on k_T (tolerance killing rate). The early transition from fast to slow killing (~12 hours post-exposure) suggests that *S. aureus* persisters do not undergo deep dormancy but rather rely on temporary metabolic shifts to withstand antibiotic pressure [6,7]. This observation supports previous findings that *S. aureus* persisters exhibit transient tolerance rather than long-term persistence, which contributes to the recurrence of infections after incomplete eradication [8,9]. Consequently, the therapeutic approach for *S. aureus* should focus on targeting early tolerance mechanisms, as persisters can be eliminated more effectively before transitioning into a protective metabolic state [10].

4.2 Model Fitting and Clinical Implications

The biphasic killing model successfully captured the persistence dynamics of both pathogens, providing quantitative evidence of their distinct survival strategies. In *M. tuberculosis*, the persistence fraction (~3.58%) was consistent with prior research on mycobacterial dormancy, where a subset of bacterial cells maintain viability for extended periods [11,12]. This finding underscores the importance of combination therapy in tuberculosis treatment, where drugs such as bedaquiline and rifampin synergistically target both actively growing and dormant subpopulations [13]. The significantly lower k_{slow} of *M. tuberculosis* (approximately ten times lower than *S. aureus*) further explains why short-term antibiotic therapy is ineffective for tuberculosis, necessitating prolonged regimens [14].

Conversely, model fitting for *S. aureus* revealed a rapid persistence decline once tolerance mechanisms were exhausted, with a near-zero dormant fraction (N_d). This suggests that tolerance-based persistence, rather than true dormancy, dictates its survival, confirming previous observations that tolerant *S. aureus* subpopulations are metabolically active and eventually cleared [15,16]. These insights highlight the potential efficacy of short, high-dose treatment regimens, which could prevent persistence-related recurrences if applied early in infection [17].

The clinical implications of these findings are profound. In *M. tuberculosis*, a metabolic reactivation strategy—involving drugs that stimulate dormant cells into a replicative state—could enhance their susceptibility to antibiotics, shortening treatment duration. In *S. aureus*, targeting early-stage persisters with metabolic inhibitors could prevent the establishment of tolerance-driven persistence, reducing the risk of chronic infection. These strategies align with current research efforts focused on developing persister-targeting compounds to complement conventional antibiotic therapy [18,19].

4.3 Future Directions

To further refine persistence modeling and enhance its translational potential, future research should focus on incorporating host-immune interactions into computational models. The immune system plays a crucial role in clearing persisters, and its interactions with antibiotic-treated bacteria

remain poorly characterized in vivo [20]. Incorporating immune-mediated clearance into persistence models could provide a more realistic prediction of bacterial survival dynamics and help design host-targeted therapies to complement antibiotics.

Additionally, single-cell transcriptomics and live-cell imaging could be integrated to validate computational predictions. These technologies enable real-time monitoring of bacterial metabolic states, providing direct experimental evidence of persistence dynamics [21]. Furthermore, machine learning-based optimization of biphasic models could refine parameter estimation and identify novel persistence-targeting drug combinations [22,23].

5. Conclusion

This study provides a quantitative and mechanistic framework for understanding bacterial persistence, offering new perspectives on optimizing treatment regimens for chronic infections. By integrating computational modeling with experimental validation, we demonstrate that *M. tuberculosis* persistence is driven by slow metabolic dormancy, requiring long-term antibiotic exposure for complete eradication. In contrast, *S. aureus* persistence is transient and tolerance-mediated, suggesting that early aggressive therapy may prevent relapses. These species-specific survival mechanisms underscore the need for precision antimicrobial strategies that go beyond traditional antibiotic regimens.

The novelty of this study lies in its quantitative characterization of persistence parameters, providing direct clinical relevance for antibiotic regimen design. The sensitivity analysis, which identifies persistence-controlling parameters, offers a new strategy for rational drug development, particularly in addressing drug-resistant tuberculosis and persistent *S. aureus* infections. These insights align with ongoing research efforts to develop novel persister-targeting compounds that complement traditional antibiotics [24,25].

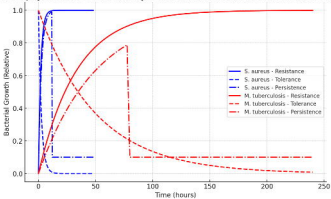
Given the global health burden of persistent infections, these findings are crucial for improving therapeutic strategies and reducing treatment durations. The integration of computational modeling into antibiotic regimen design represents a major advancement in precision medicine, providing a predictive framework for optimizing bacterial eradication. Future research should expand upon these models by incorporating immune system interactions, patient-specific bacterial profiles, and novel persistence-targeting drugs. By leveraging computational tools alongside experimental validation, we can accelerate the development of more effective, personalized treatments, ultimately reducing the burden of chronic and drug-resistant bacterial infections [26-30].

6. References

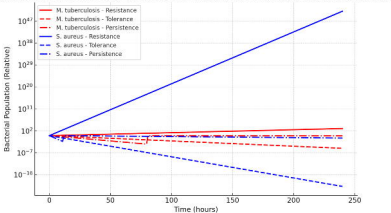
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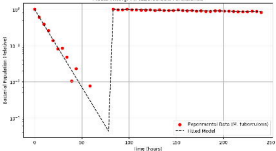
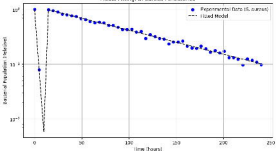
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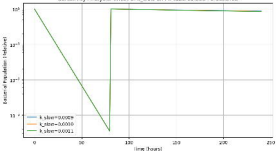
Comparison of Staphylococcus aureus and Mycobacterium tuberculosis Growth Under Antibiotic Exposure



Simulation of Resistance, Tolerance, and Persistence in *M. tuberculosis* vs. *S. aureus*



Model Fitting: *M. tuberculosis* PersistenceModel Fitting: *S. aureus* Persistence

Sensitivity Analysis: Effect of k_{slow} on *M. tuberculosis* PersistenceSensitivity Analysis: Effect of k_{F} on *S. aureus* Persistence